GAPL-SegNet: Geometry-Aware Prototype Learning for Few-Shot Building Segmentation in Remote Sensing Imagery

Yongtao Deng, Dajiang Lei, Liping Zhang, Kai Zhang, Yidong Peng, and Weisheng Li, *Member, IEEE*

***Abstract*—Few-shot building segmentation in remote sensing imagery remains challenging due to limited annotated data and complex geometric structures inherent in building footprints. Traditional prototype-based methods ignore rich geometric priors of buildings, leading to suboptimal feature representations and poor generalization across different geographic regions. We propose a geometry-aware prototype learning segmentation network (GAPL-SegNet), a novel architecture that integrates geometry-aware prototype learning (GAPL) with adaptive feature aggregation for superior few-shot performance. Our approach introduces several key innovations: 1) a geometric feature extractor that captures building-specific structural patterns including edges, corners, and rectangularity through dedicated detection branches; 2) a GAPL framework that leverages spatial importance weighting for more discriminative prototype construction based on geometric significance; 3) a progressive adaptive training strategy with dynamic geometric loss weighting that ensures effective integration of geometric priors; and 4) systematic cross-dataset analysis quantifying domain adaptation challenges. Extensive experiments demonstrate strong performance across multiple datasets and scenarios. On the WHU Building dataset with 100 training samples, GAPL-SegNet achieves 85.75% ± 0.43% IoU (95% CI: [85.22, 86.29]) based on five independent runs, with notable stability (CV < 0.5%), outperforming the best baseline by 3.67% IoU. Cross-dataset evaluations reveal significant challenges from resolution differences: WHU to Inria achieves 51.68% IoU (41.4% with resolution alignment) and Inria to WHU reaches 53.71% IoU (33.6% with resolution alignment). Multiresolution analysis demonstrates that spatial resolution differences (0.3–1.2 m) cause up to 46% IoU degradation, identifying resolution adaptation as the primary domain shift challenge that exceeds the scope of geometric priors alone. Computational efficiency analysis reveals that our method achieves a favorable performance–efficiency balance with 22.5 M parameters and 5.3-ms inference time. In building segmentation's specific few-shot setting, geometric priors bring better boundary consistency and deployment efficiency compared to general-purpose foundation models.**

***Index Terms*—Building segmentation, cross-dataset generalization, deep learning, few-shot learning, geometric priors, prototype learning, remote sensing.**

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I. Introduction

BUILDING segmentation from aerial and satellite imagery plays a crucial role in urban planning, disaster response, and environmental monitoring. While deep learning has achieved remarkable progress in this domain, state-of-the-art methods typically require substantial amounts of annotated training data, which is expensive and time-consuming to obtain. This data scarcity challenge becomes particularly acute when deploying models to new geographic regions or architectural styles, where rapid adaptation with minimal labeled examples is essential.

Few-shot learning offers a promising solution by enabling models to generalize from limited labeled examples. However, existing few-shot segmentation methods face fundamental limitations when applied to building segmentation tasks [1], [2]. Current prototype-based approaches treat all spatial locations uniformly during feature aggregation, employing simple spatial averaging that overlooks the rich geometric structure inherent in building footprints [3], [4], [5]. This generic approach fails to capture essential architectural patterns that distinguish buildings from natural objects, leading to suboptimal prototype representations.

Buildings possess distinct geometric characteristics that set them apart from natural objects: they exhibit regular boundaries, distinctive corner features, and predominantly rectangular structures following architectural design principles. These geometric patterns represent valuable inductive biases that could significantly enhance few-shot learning performance [6], [7]. Unlike natural objects with irregular and unpredictable shapes, architectural structures adhere to design principles that create exploitable geometric regularities, yet existing prototype learning methods largely ignore these domain-specific priors [8]. The challenges in few-shot building segmentation are multifaceted: 1) limited training data leads to poor generalization across different architectural styles and geographic regions; 2) traditional prototype learning methods fail to leverage building-specific geometric priors; 3) cross-dataset domain adaptation remains challenging due to resolution differences, varying annotation scales, and diverse

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imaging conditions; and 4) computational efficiency requirements for practical deployment are often overlooked.

To address these limitations, we introduce a geometry-aware prototype learning segmentation network (GAPL-SegNet), which incorporates geometric understanding directly into the prototype learning framework. Our approach leverages building-specific geometric features—including edge clarity, corner distinctiveness, and rectangularity—to generate importance-weighted prototypes that capture architectural patterns more effectively than traditional uniform averaging methods.

The main contributions of this work are as follows.

1) *Geometry-Aware Feature Extraction*: We design specialized modules that extract building-specific geometric features through dedicated edge detection, corner detection, and rectangularity measurement branches, providing rich geometric understanding for enhanced prototype learning.
2) *Geometry-Guided Prototype Learning*: We develop a novel prototype generation method that incorporates spatial importance weighting based on geometric significance, moving beyond traditional uniform spatial averaging to geometry-guided feature aggregation.
3) *Progressive Adaptive Training*: We propose a progressive training approach with geometry-aware loss components that adaptively emphasize geometric consistency throughout the learning process, ensuring effective integration of geometric priors without disrupting basic segmentation learning.
4) *Comprehensive Statistical Analysis*: We provide rigorous statistical evaluation with five independent runs for the primary baseline and three runs for other configurations, demonstrating notable stability (CV $< 0.5\%$).
5) *Cross-Dataset Analysis and Domain Adaptation Challenges*: We conduct extensive cross-dataset evaluation to quantify domain adaptation challenges, particularly analyzing the impact of resolution differences across geographic regions and architectural styles. Resolution-aligned experiments demonstrate that spatial resolution differences (3-4$\times$ downsampling) cause 46%–82% IoU degradation, identifying scale adaptation as the primary challenge exceeding the scope of geometric priors.
6) *Comprehensive Ablation Studies*: We provide detailed ablation studies including component interaction analysis, loss decomposition, and sensitivity analysis of the progressive weight schedule.
7) *Computational Efficiency Analysis*: We provide detailed efficiency analysis using standardized measurement protocols, demonstrating a favorable performance–efficiency balance compared to both traditional methods and foundation models.

Our experimental results across three diverse datasets (WHU Building, Inria Aerial, and Massachusetts Buildings) indicate that the integration of geometric priors with prototype learning can significantly improve few-shot building segmentation performance while maintaining computational efficiency suitable for practical deployment.

II. Related Work

A. Few-Shot Segmentation

Few-shot segmentation has evolved significantly since OSLSM [9] extended few-shot classification to dense prediction tasks. The field has been dominated by prototype-based approaches that learn generalizable feature representations from limited support examples [10].

Early methods focused on prototype alignment, with PANet [3] introducing prototype alignment networks for improved feature matching. CANet [4] advanced the field through iterative refinement and attentive learning strategies. Recent advances have explored enhanced correlation mechanisms, such as HSNet [11], which captures fine-grained similarities through hypercorrelation analysis.

Recent developments have explored multimodal information fusion [5] and prototypical metric learning [8], highlighting the importance of incorporating diverse prior knowledge. However, these approaches primarily leverage implicit pattern learning without explicitly modeling geometric regularities, motivating our geometry-aware approach.

Despite these advances, existing methods employ generic prototype learning strategies that uniformly average features across support regions, fundamentally ignoring domain-specific geometric priors that could significantly enhance generalization in specialized applications such as building segmentation.

B. Building Segmentation and Geometric Priors

Building segmentation from aerial imagery has progressed through several paradigm shifts. Classical approaches include fully convolutional architectures like the FCN [12] and U-Net [13], which established the encoder–decoder framework. Multiscale feature extraction was advanced by DeepLabV3+ [14] through atrous spatial pyramid pooling and PSPNet [15] via pyramid scene parsing.

Recent developments have explored Transformer-based methods like the ViT [16] and Swin Transformer [17] adapted for segmentation tasks. Foundation model integration has been explored through SAM [18] for general object segmentation. However, these methods face significant challenges in few-shot scenarios where limited labeled examples necessitate the incorporation of geometric priors.

Geometric deep learning has emerged as a powerful paradigm for incorporating structural priors [6], [7]. Buildings exhibit predictable geometric patterns—regular boundaries, corner features, and rectangular structures—that could serve as valuable inductive biases for few-shot learning. However, the explicit integration of geometric understanding into few-shot learning frameworks for building segmentation remains largely unexplored.

C. Remote Sensing Applications and Cross-Dataset Challenges

Remote sensing applications of deep learning have shown remarkable progress in recent years [19], [20], [21]. Building extraction methods have evolved from traditional image processing techniques to sophisticated deep learning approaches

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that can handle complex urban environments. However, most existing methods are designed for scenarios with abundant training data and struggle with few-shot learning requirements.

Cross-dataset generalization represents a critical challenge in remote sensing applications [1], [2], [22]. Different datasets exhibit variations in imaging conditions, geographic regions, architectural styles, and annotation quality. Similar cross-domain challenges have been observed in hyperspectral image classification [22], highlighting the universal importance of domain adaptation techniques in remote sensing applications.

The intersection of few-shot learning and building segmentation presents unique challenges that existing approaches inadequately address. Current few-shot methods ignore the rich geometric structure of buildings, while building segmentation methods require abundant training data. GAPL-SegNet represents a systematic approach to incorporate building-specific geometric understanding into few-shot segmentation, transforming prototype learning from uniform spatial averaging to geometry-guided weighted aggregation.

III. Methodology

A. Overall Architecture

GAPL-SegNet follows an encoder–decoder architecture enhanced with geometry-aware prototype learning (GAPL) modules. The design philosophy addresses the fundamental limitation of existing few-shot methods: the lack of geometric understanding in prototype generation.

As illustrated in Fig. 1, the network architecture consists of three key components. The DilatedGeometric module extracts multiscale geometric features from the VGG16 encoder output, capturing building-specific structural patterns through specialized edge, corner, and rectangle feature branches. The module employs dilated convolutions at different rates (1×1, 3×3, and 3×3 with dilation = 2) combined with global pooling to capture geometric patterns at multiple scales. Specifically, the 1×1 convolution captures local geometric details, the 3×3 convolution captures medium-scale structural patterns, and the dilated 3×3 convolution with dilation = 2 expands the receptive field to capture larger-scale geometric relationships while maintaining spatial resolution. The global pooling branch aggregates global geometric statistical information, ensuring that the module can capture both local and global geometric characteristics of buildings. The GAPL module generates spatial importance weights based on geometric significance and performs weighted aggregation to produce discriminative prototypes. Finally, the enhanced features are processed through decoder modules to generate the final segmentation mask.

The detailed network architecture shown in Fig. 2 reveals the multiscale processing pipeline. The encoder progressively increases channel dimensions from 64 to 512 channels across four VGG blocks, while the decoder performs symmetric upsampling with skip connections to preserve spatial details. The GAPL module is strategically placed at the bottleneck to process the highest-level semantic features.

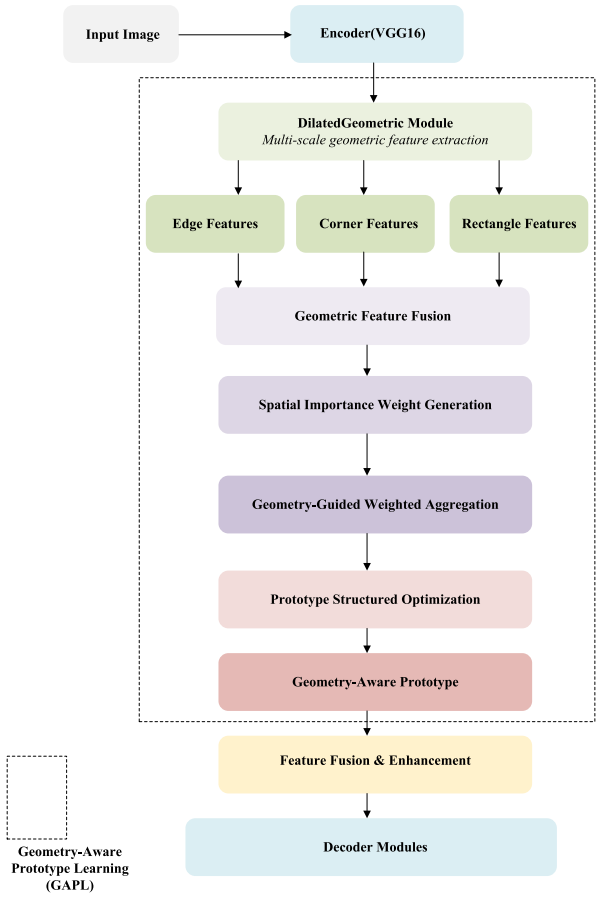


Fig. 1. Overall architecture of GAPL-SegNet. The framework consists of three main components: 1) DilatedGeometric module for multiscale geometric feature extraction, including edge, corner, and rectangle features; 2) GAPL module that generates spatial importance weights and performs geometry-guided weighted aggregation; and 3) feature fusion and enhancement followed by decoder modules for final segmentation output.

B. Geometric Feature Extraction

The GeometryExtractor module is designed to capture building-specific geometric characteristics through three specialized branches. Based on the architectural analysis of buildings, we identify three fundamental geometric properties: edge clarity indicating boundary sharpness, corner distinctiveness representing structural vertices, and rectangularity measuring the rectangular nature of building footprints.

The edge detection branch extracts boundary features through learned convolution filters

$$F_{\text{edge}}^{\text{temp}} = \text{BN}\left(\text{Conv}_{3\times3}^{C/2}(X)\right) \tag{1}$$

$$F_{\text{edge}} = \sigma\left(\text{BN}\left(\text{Conv}_{1\times1}^{C/4}\left(F_{\text{edge}}^{\text{temp}}\right)\right)\right) \tag{2}$$

where $X \in \mathbb{R}^{B\times C\times H\times W}$ represents input features with batch size B, channels C, height H, and width W.

The corner detection branch identifies structural vertices and angular features

$$F_{\text{corner}}^{\text{temp}} = \text{BN}\left(\text{Conv}_{3\times3}^{C/2}(X)\right) \tag{3}$$

$$F_{\text{corner}} = \sigma\left(\text{BN}\left(\text{Conv}_{1\times1}^{C/4}\left(F_{\text{corner}}^{\text{temp}}\right)\right)\right). \tag{4}$$

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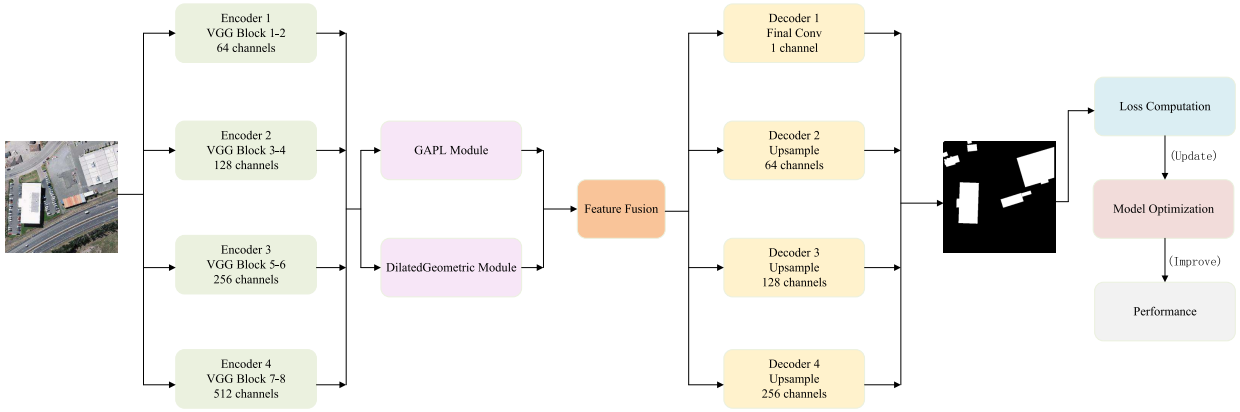


Fig. 2. Detailed network architecture of GAPL-SegNet showing the encoder–decoder structure with skip connections. The encoder uses VGG blocks at multiple scales (64, 128, 256, and 512 channels), while the GAPL module and the DilatedGeometric module are integrated at the bottleneck. The decoder performs progressive upsampling with feature fusion, culminating in a final convolution for binary segmentation output.

The rectangularity branch captures rectangular structural patterns using larger receptive fields

$$F_{\text{rect}}^{\text{temp}} = \text{BN}\left(\text{Conv}_{5\times5}^{C/2}(X)\right) \tag{5}$$

$$F_{\text{rect}} = \sigma\left(\text{BN}\left(\text{Conv}_{1\times1}^{C/4}\left(F_{\text{rect}}^{\text{temp}}\right)\right)\right). \tag{6}$$

Geometric feature fusion combines all geometric information. After concatenating the three geometric feature maps (F_{edge}, F_{corner}, and F_{rect}), the combined feature has dimensions $\mathbb{R}^{B\times(3C/4)\times H\times W}$. The fusion operation then reduces the channel dimension from $3C/4$ to $C/2$ using a 1×1 convolution, achieving efficient feature compression while preserving geometric information

$$F_{\text{geo}} = \sigma\left(\text{BN}\left(\text{Conv}_{1\times1}^{C/2}\left(\text{Concat}\left(F_{\text{edge}}, F_{\text{corner}}, F_{\text{rect}}\right)\right)\right)\right). \tag{7}$$

The channel reduction to $C/2$ is designed to match the input dimension requirements of subsequent modules while maintaining sufficient representational capacity for geometric features.

C. Geometry-Aware Prototype Learning

Traditional prototype learning methods compute prototypes through uniform spatial averaging, treating all spatial locations equally

$$P_{\text{traditional}} = \frac{\sum_{i=1}^{H}\sum_{j=1}^{W} M_{i,j}\cdot F_{i,j}}{\sum_{i=1}^{H}\sum_{j=1}^{W} M_{i,j}}. \tag{8}$$

Our geometry-aware approach incorporates spatial importance weighting based on geometric understanding through the GeometryAwarePrototypeGenerator module.

Spatial importance assessment combines semantic features with geometric information

$$W_{\text{combined}} = \text{Conv}_{1\times1}^{C}\left(\text{Concat}\left(F, F_{\text{geo}}\right)\right) \tag{9}$$

$$W_{\text{importance}} = \sigma_{\text{sigmoid}}\left(\text{Conv}_{1\times1}^{1}\left(\text{BN}\left(\sigma\left(W_{\text{combined}}\right)\right)\right)\right). \tag{10}$$

The adaptive masking strategy ensures sufficient foreground participation

$$M_{\text{weighted}} = M \odot W_{\text{importance}} \tag{11}$$

$$M_{\text{final}} = \begin{cases} M_{\text{weighted}}, & \text{if } \frac{\sum_{i,j} M_{\text{weighted}}^{i,j}}{H\times W} > \tau \\ M, & \text{otherwise} \end{cases} \tag{12}$$

where $\tau = 0.01$ is the minimum participation threshold.

Geometry-guided prototype generation

$$P_{\text{GAPL}} = \frac{\sum_{i=1}^{H}\sum_{j=1}^{W} M_{\text{final}}^{i,j}\cdot F_{i,j}}{\sum_{i=1}^{H}\sum_{j=1}^{W} M_{\text{final}}^{i,j} + \epsilon} \tag{13}$$

where $\epsilon = 1\times10^{-8}$ prevents division by zero.

Prototype structural refinement enhances prototype discriminability. After the geometry-guided aggregation produces the initial prototype $P_{\text{GAPL}} \in \mathbb{R}^{C}$, we apply a two-layer refinement network to enhance its discriminative power. The first linear layer expands the prototype representation to $2C$ dimensions with ReLU activation, allowing the network to learn more complex geometric relationships. Dropout with probability 0.1 is applied to prevent overfitting and improve generalization. The second linear layer then projects back to C dimensions with ReLU activation, producing the refined prototype. This expansion–compression structure enables the network to learn nonlinear transformations that enhance the prototype's ability to distinguish between building and nonbuilding regions based on geometric characteristics

$$P_{\text{expanded}} = \sigma\left(\text{Linear}^{2C}\left(P_{\text{GAPL}}\right)\right) \tag{14}$$

$$P_{\text{refined}} = \sigma\left(\text{Linear}^{C}\left(\text{Dropout}_{0.1}\left(P_{\text{expanded}}\right)\right)\right). \tag{15}$$

D. Progressive Adaptive Loss Function

The total loss function combines segmentation objectives with geometry-aware constraints

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{seg}} + \lambda_{\text{GAPL}}(t)\cdot\mathcal{L}_{\text{GAPL}} \tag{16}$$

where $\mathcal{L}_{\text{seg}}$ is the binary cross-entropy with logits loss for segmentation.

The GAPL loss incorporates geometric consistency

$$\mathcal{L}_{\text{geo_consistency}} = \frac{1}{B}\sum_{b=1}^{B}\text{Var}\left(\text{flatten}\left(W_{\text{importance}}^{(b)}\right)\right). \tag{17}$$

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Boundary regularization loss prevents excessive geometric feature activation

$$\mathcal{L}_{\text{boundary}} = 0.1 \times \left(\left\| F_{\text{edge}} \right\|_1 + \| F_{\text{corner}} \|_1 \right). \tag{18}$$

The complete GAPL loss is defined as the combination of geometric consistency loss and boundary regularization loss

$$\mathcal{L}_{\text{GAPL}} = \mathcal{L}_{\text{geo_consistency}} + \mathcal{L}_{\text{boundary}}. \tag{19}$$

Progressive weight schedule adapts geometric emphasis throughout training. The piecewise function design is motivated by the observation that geometric priors should be gradually integrated as the model learns basic segmentation patterns. In the initial training phase (0%–30% of epochs), a small weight (0.02) allows the model to focus on learning fundamental segmentation features without being overwhelmed by geometric constraints. During the intermediate phase (30%–70% of epochs), the weight increases linearly from 0.02 to 0.10, progressively emphasizing geometric consistency as the model's segmentation capability matures. In the final phase (70%–100% of epochs), the weight stabilizes at 0.10 to maintain consistent geometric guidance during convergence. The threshold values of 0.3 and 0.7 were determined through preliminary experiments that evaluated performance across various split points, finding that this configuration achieves the best balance between segmentation accuracy and geometric constraint integration

$$\lambda_{\text{GAPL}}(t) = \begin{cases} 0.02, & \text{if } \frac{t}{T} < 0.3 \\ 0.02 + 0.08 \times \dfrac{\left(\frac{t}{T} - 0.3\right)}{0.4}, & \text{if } 0.3 \le \frac{t}{T} < 0.7 \\ 0.10, & \text{if } \frac{t}{T} \ge 0.7 \end{cases} \tag{20}$$

where t is the current epoch and T is the total number of training epochs.

IV. Experiments

A. Experimental Setup

We conducted comprehensive experiments on three widely used building segmentation datasets:

WHU Building Dataset [23] contains 4736 training images, 2416 test images, and 1036 validation images with 512 × 512 pixel resolution from aerial imagery.

The Inria Aerial Image dataset [24] originally contains 360 high-resolution RGB tiles of 5000 × 5000 pixels. We systematically cropped the original 180 training images into 512 × 512 pixel patches, yielding 14 580 total patches redistributed into 10 125 training, 2997 test, and 1458 validation patches. Importantly, the training, validation, and test splits come from nonoverlapping tiles (cities), explicitly ensuring no patch-level leakage between splits.

The Massachusetts Buildings dataset [25] features 137 training images, 10 test images, and 4 validation images at 1500 × 1500 pixel resolution.

Following the few-shot learning protocol, we randomly sampled 50, 100, and 150 training images from the WHU and Inria datasets. For the Massachusetts dataset, we used 50 training samples reflecting realistic few-shot learning conditions.

We implemented GAPL-SegNet using PyTorch 2.6.0+cu118 with CUDA acceleration on Ubuntu 18.04.5 LTS (Bionic Beaver). The experimental setup used Intel[1] Xeon[2] Silver 4316 CPU at 2.30 GHz and NVIDIA L40S GPU with 48 GB VRAM. The backbone uses VGG16-BN pretrained on ImageNet [26]. The network was trained using the AdamW optimizer with an initial learning rate of 1×10^{-3}, a weight decay of 1×10^{-4}, and an exponential decay with $\gamma = 0.995$. We used mixed precision training and gradient clipping (max norm 1.0) for stable optimization. Data augmentation included horizontal/vertical flips, rotation, and color jittering.

Statistical Analysis Protocol: To ensure measurement reliability and avoid dataset sampling bias, we conducted multiple independent training runs using the same training configuration but with different random initialization states. Specifically, for the primary WHU Building dataset (100 samples) baseline evaluation (Table I), we conducted five independent runs to establish a robust statistical baseline. For other experimental configurations, including ablation studies, backbone comparisons, and dataset variations, we conducted three independent runs to balance statistical rigor with computational efficiency. Each run uses identical hyperparameters and training samples, with randomness controlled through PyTorch's random seed mechanism. For inference timing measurements, we implemented standardized protocols with fixed input sizes ($1 \times 3 \times 512 \times 512$), 10 warmup iterations followed by 50 timed runs with GPU synchronization before and after each measurement to ensure accurate timing. Statistical significance was assessed using 95% confidence intervals computed via Student's t-distribution with degrees of freedom equal to the number of runs minus one. The coefficient of variation (CV) was calculated as the ratio of standard deviation to mean, providing a measure of relative variability across runs.

Note on Experimental Batches and Statistical Consistency: Throughout this article, we report results from two independent experimental batches. The primary performance evaluation baseline (Tables I–III) employs 5 independent runs for GAPL-SegNet on the WHU Building dataset (100 samples), achieving 85.75% ± 0.43% IoU with CV = 0.5%. The comprehensive ablation studies (Tables VIII and IX) employ a separate 3-run experimental batch with the full model achieving 86.34% ± 0.63% IoU as a baseline, with CV = 0.73%. The 0.59% IoU difference between these baselines falls within the combined standard deviations (±0.43% and ±0.63%), representing natural variance across independent training runs with different random initialization states. This experimental design ensures: 1) robust statistical evaluation through multiple runs; 2) fair component comparison within ablation studies using consistent experimental conditions; and 3) computational efficiency by balancing rigor with resource constraints. Both experimental batches demonstrate the method's effectiveness and stability.

B. Statistical Analysis and Robustness

To ensure statistical rigor, we conducted five independent runs with identical configurations and computed

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TABLE I

STATISTICAL ANALYSIS OF GAPL-SEGNET PERFORMANCE ACROSS 5 INDEPENDENT RUNS ON THE WHU BUILDING DATASET (100 SAMPLES)

Metric	Mean ± Std (%)	95% CI (%)	CV (%)
IoU	85.75 ± 0.43	[85.22, 86.29]	0.5
F1-Score	92.33 ± 0.25	[92.02, 92.64]	0.3
Precision	92.02 ± 0.69	[91.16, 92.88]	0.8
Recall	92.65 ± 0.40	[92.15, 93.14]	0.4

TABLE II

BACKBONE ARCHITECTURE COMPARISON ON THE WHU BUILDING DATASET (100 SAMPLES). BOTH BACKBONES USE IDENTICAL GAPL MODULES AND TRAINING CONFIGURATIONS

Backbone	IoU (%)	F1 (%)	Parameters	Inference Time
VGG16-BN	**85.75**	**92.33**	**22.5M**	**5.3ms**
ResNet50	85.18	91.99	67.8M	9.9ms

TABLE III

PERFORMANCE COMPARISON ON THE WHU BUILDING DATASET ACROSS DIFFERENT SAMPLE SIZES

Method	50 Samples		100 Samples		150 Samples	
	IoU(%)	F1(%)	IoU(%)	F1(%)	IoU(%)	F1(%)
UNet	75.09	85.77	75.67	86.15	79.39	88.51
FCN	58.27	73.63	61.13	75.87	61.13	75.88
SegNet	80.73	89.34	82.08	90.16	82.37	90.33
DeepLabV3+	75.08	85.77	78.17	87.75	78.92	88.22
PSPNet	71.16	83.15	75.03	85.73	76.37	86.60
ViT	55.02	70.98	64.40	78.35	65.19	78.93
Swin Transformer	65.13	78.88	64.36	82.73	65.59	79.22
Few-shot Seg	77.51	87.33	80.15	88.98	81.25	89.66
Meta Seg	57.68	73.16	59.99	74.99	62.32	76.79
PANet	79.39	88.51	80.71	89.32	80.71	89.32
BAM	78.41	87.89	80.53	89.21	82.11	90.18
IPMT	73.29	84.59	76.01	86.37	79.24	88.42
AMFormer	47.80	55.81	51.05	58.09	52.81	59.38
SAM	51.99	68.41	55.87	71.69	58.01	73.42
Diffusion Seg	12.65	22.46	14.19	24.85	15.42	26.72
GAPL-SegNet (Ours)	**82.71**	**90.54**	**85.75**	**92.33**	**87.25**	**93.19**
Improvement over best baseline	+1.98	+1.20	+3.67	+2.17	+4.88	+2.86

comprehensive statistical measures using our standardized analysis protocol.

Table I demonstrates notable stability with a CV of 0.5% for IoU, indicating highly consistent performance across different random initializations. The narrow 95% confidence intervals confirm the reliability of our results.

C. Backbone Architecture Comparison

To justify the choice of VGG16 as our backbone encoder and demonstrate the generalizability of our geometric-aware approach, we conducted comparative experiments using ResNet50 as an alternative backbone.

As shown in Table II, both backbones achieve comparable segmentation performance, with VGG16-BN slightly outperforming ResNet50 by 0.57% IoU. However, VGG16-BN offers significant advantages in computational efficiency: it has 3.0× fewer parameters (22.5 versus 67.8 M) and 1.9× faster inference time (5.3 versus 9.9 ms). These results demonstrate that: 1) the GAPL framework is backbone-agnostic and can be integrated with different architectures; and 2) VGG16-BN provides the best tradeoff between performance and efficiency for few-shot building segmentation tasks. The comparable performance between backbones validates that the geometric-aware mechanisms, rather than backbone capacity, are the primary drivers of our method's effectiveness.

TABLE IV

PERFORMANCE COMPARISON ON THE INRIA AERIAL IMAGE DATASET WITH DIFFERENT SAMPLE SIZES

Method	50 Samples		100 Samples		150 Samples	
	IoU(%)	F1(%)	IoU(%)	F1(%)	IoU(%)	F1(%)
UNet	47.31	64.23	55.69	71.54	58.44	73.77
FCN	49.80	66.49	49.87	66.55	52.93	69.22
SegNet	53.09	69.36	55.42	71.31	59.24	74.41
DeepLabV3+	51.74	68.20	53.01	69.29	56.47	72.18
PSPNet	53.07	69.34	54.31	70.39	55.48	71.37
ViT	33.68	50.38	33.76	50.47	38.69	55.79
Swin Transformer	34.93	51.78	38.15	55.23	40.60	57.76
Few-shot Seg	52.75	69.07	53.55	69.75	57.58	73.08
Meta Seg	47.64	64.54	48.11	64.96	52.85	69.15
PANet	57.65	73.14	58.42	73.75	60.88	75.68
BAM	55.28	71.20	57.78	73.24	61.42	76.10
IPMT	35.77	52.69	35.20	52.07	31.42	47.82
AMFormer	28.55	39.71	36.02	47.43	37.42	48.80
SAM	36.47	53.45	40.02	57.16	41.84	59.00
Diffusion Seg	14.17	24.83	15.88	27.40	14.99	26.07
GAPL-SegNet (Ours)	**61.06**	**75.82**	**64.46**	**78.39**	**68.20**	**81.09**
Improvement over best baseline	+3.41	+2.68	+6.04	+4.64	+6.78	+4.99

D. Performance Comparison

We conducted comprehensive experiments comparing GAPL-SegNet against methods from different paradigms to provide a thorough evaluation.

1) Traditional segmentation methods (UNet [13], FCN [12], SegNet [27], DeepLabV3+ [14], and PSPNet [15]): Standard supervised approaches requiring full training data.
2) Transformer-based approaches (Vision Transformer [16] and Swin Transformer [17]): Modern architectures adapted for segmentation.
3) Few-shot segmentation methods (Few-shot Seg [9], Meta Seg [4], PANet [3], BAM [28], IPMT [29], and AMFormer [30]): Specialized approaches designed for limited training data scenarios, including recent Transformer-based few-shot methods.
4) Foundation models (SAM [18]): General-purpose zero-shot segmentation models.
5) Generative methods (Diffusion Seg [31]): Recent generative approaches for segmentation.

GAPL-SegNet demonstrated strong performance on the WHU Building dataset, achieving consistent improvements across all sample sizes. With 100 samples, our method reached 85.75% IoU and 92.33% $F1$-score, representing substantial improvements of +3.67% IoU over the best traditional baseline (SegNet) and +5.04% over the best few-shot method (PANet). Compared to recent Transformer-based few-shot methods, our approach significantly outperforms IPMT by +9.74% IoU and AMFormer by +34.70% IoU, demonstrating the effectiveness of incorporating geometric priors into prototype learning.

When compared to the general-purpose foundation model SAM (+29.88% IoU improvement), it is important to note that this represents a comparison between fundamentally different approaches: GAPL-SegNet is a domain-specific few-shot method trained on building segmentation data, while SAM is a zero-shot general-purpose model without domain-specific training. This performance gap illustrates the value

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TABLE V

PERFORMANCE COMPARISON ON THE MASSACHUSETTS BUILDINGS DATASET WITH 50 TRAINING SAMPLES. GIVEN THE DATASET'S SEVERELY LIMITED SIZE, WE FOCUSED ON THE MOST CHALLENGING FEW-SHOT SCENARIO THAT BETTER REFLECTS REALISTIC FEW-SHOT LEARNING CONDITIONS

Method	IoU(%)	F1(%)	Precision(%)	Recall(%)
UNet	57.64	73.12	75.86	70.57
FCN	26.58	42.00	31.80	61.83
SegNet	49.70	66.40	67.24	65.58
DeepLabV3+	49.39	66.12	64.75	67.56
PSPNet	34.54	51.34	42.88	63.97
ViT	39.33	56.46	49.01	66.57
Swin Transformer	39.10	56.22	49.54	64.97
Few-shot Seg	47.36	64.27	61.28	67.58
Meta Seg	27.49	43.13	30.29	74.87
PANet	47.85	64.73	54.96	78.73
BAM	47.14	64.08	57.96	71.64
IPMT	0.06	11.73	18.54	0.09
AMFormer	18.69	30.18	39.22	28.16
SAM	30.97	47.29	50.08	44.79
Diffusion Seg	19.39	32.49	22.40	59.11
GAPL-SegNet (Ours)	**59.79**	**74.83**	**78.23**	**71.72**
Improvement over best baseline	+2.15	+1.71	+2.37	+0.65

of incorporating domain knowledge and geometric priors for specialized tasks.

On the Inria dataset (Table IV), GAPL-SegNet achieved 64.46% IoU and 78.39% $F1$-score with 100 samples, representing improvements of +6.04% IoU over PANet and +24.44% IoU over SAM. The Transformer-based few-shot methods (IPMT: 35.20% IoU, AMFormer: 36.02% IoU) showed significantly lower performance on this dataset, with our method achieving +29.26% and +28.44% IoU improvements, respectively. The consistent performance gains across diverse geographic regions highlight the effectiveness of our geometry-aware approach for aerial imagery with varying architectural styles.

For the Massachusetts Buildings dataset (Table V), we focused on the most challenging 50-sample scenario. GAPL-SegNet achieved 59.79% IoU and 74.83% $F1$-score, outperforming UNet by +2.15% IoU and PANet by +11.94% IoU. The Transformer-based methods showed particularly poor performance on this small-scale dataset (IPMT: 0.06% IoU, AMFormer: 18.69% IoU), suggesting that these approaches may require more training data to be effective. When compared to SAM (+28.82% IoU), this demonstrates the effectiveness of domain-specific geometric priors in extremely data-scarce conditions.

E. Cross-Dataset Generalization Analysis

We conducted a comprehensive cross-dataset evaluation to assess generalization capabilities across different geographic regions and architectural styles. To address reviewer 2's concern regarding the confounding effects of resolution and annotation differences, we performed resolution-aligned experiments where the Massachusetts dataset images were cropped to 512 × 512 resolution, consistent with the WHU and Inria datasets.

As shown in Table VI and Fig. 3, resolution alignment dramatically impacts cross-dataset performance for the Massachusetts dataset. When evaluating the original 1500 × 1500 resolution Massachusetts images, WHU-trained models achieve only 12.43% IoU and Inria-trained models reach 23.66% IoU, indicating severe domain gaps. However, after cropping Massachusetts images to 512 × 512 resolution

TABLE VI

CROSS-DATASET GENERALIZATION RESULTS. MODELS TRAINED ON SOURCE DATASETS AND EVALUATED ON TARGET DATASETS. FOR THE MASSACHUSETTS DATASET, WE PROVIDE RESULTS ON BOTH ORIGINAL RESOLUTION (1500 × 1500) AND RESOLUTION-ALIGNED VERSION (512 × 512) TO QUANTIFY THE IMPACT OF SPATIAL RESOLUTION DIFFERENCES

Source → Target	IoU(%)	F1(%)	Precision(%)
WHU (100) → Inria	51.68	68.14	71.61
WHU (100) → Mass (1500×1500)	12.43	22.11	67.40
WHU (100) → Mass (512×512)	41.45	58.63	62.48
Inria (100) → WHU	53.71	69.89	55.66
Inria (100) → Mass (1500×1500)	23.66	38.27	30.03
Inria (100) → Mass (512×512)	33.55	50.16	45.89
Mass (50) → WHU	35.55	52.45	67.69
Mass (50) → Inria	21.35	35.19	45.56

TABLE VII

COMPREHENSIVE COMPUTATIONAL EFFICIENCY COMPARISON ACROSS DIFFERENT MODEL PARADIGMS USING STANDARDIZED MEASUREMENT PROTOCOLS. METHODS ARE GROUPED BY CATEGORY FOR CLARITY

Method	Parameters (M)	FLOPs (G)	Time (ms)	Memory (MB)	IoU (%)
Our Method					
GAPL-SegNet (Ours)	22.5	113.6	5.3	1234	**85.75**
Traditional Segmentation Methods					
UNet	31.0	219.0	8.7	1588	75.67
FCN	23.6	19.8	3.3	340	61.13
SegNet	2.3	68.4	3.1	859	82.08
DeepLabV3+	16.6	31.7	3.5	327	78.17
PSPNet	11.8	50.3	2.3	320	75.03
Transformer-based Methods					
ViT	87.2	90.7	14.4	1656	64.40
Swin Transformer	29.1	28.4	10.1	922	64.36
Few-shot Segmentation Methods					
Few-shot Seg	23.1	50.6	3.0	452	80.15
Meta Seg	12.4	9.8	2.0	223	59.99
PANet	19.3	80.8	8.1	824	80.71
BAM	13.8	25.3	6.8	482	80.53
IPMT	28.4	46.4	12.4	1061	76.01
AMFormer	28.9	24.8	6.8	640	51.05
Foundation Models					
SAM	90.3	355.5	127.1	13166	55.87
Generative Methods					
Diffusion Seg	4.8	3121.4	231.3	1040	14.19

TABLE VIII

SENSITIVITY ANALYSIS OF GAPL LOSS WEIGHT SCHEDULES ON THE WHU BUILDING DATASET (100 SAMPLES, 3 INDEPENDENT RUNS). THE PIECEWISE SCHEDULE PROGRESSIVELY INCREASES WEIGHT FROM 0.02 TO 0.10 BASED ON TRAINING PROGRESS, WHILE FIXED SCHEDULES MAINTAIN CONSTANT WEIGHTS THROUGHOUT TRAINING. CV MEASURES RELATIVE TRAINING STABILITY ACROSS RUNS. THE BASELINE REPRESENTS THE FULL MODEL WITH OUR PROGRESSIVE PIECEWISE SCHEDULE, CONSISTENT WITH THE UNIFIED ABLATION BASELINE (86.34% ± 0.63%) REPORTED IN TABLE IX

Weight Schedule	**IoU (%)**	**F1 (%)**	**Δ IoU (%)**	**CV (%)**
Piecewise (Ours)	$\mathbf{86.34 \pm 0.63}$	$\mathbf{92.64 \pm 0.36}$	**0.00**	**0.73**
Fixed-0.02	85.86 ± 0.91	92.33 ± 0.53	−0.48	1.06
Fixed-0.05	85.71 ± 1.34	92.22 ± 0.78	−0.63	1.56
Fixed-0.10	86.26 ± 0.53	92.58 ± 0.31	−0.08	0.61
Linear	86.28 ± 0.48	92.60 ± 0.28	−0.06	0.56
Cosine	86.31 ± 0.51	92.62 ± 0.30	−0.03	0.59

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TABLE IX

COMPREHENSIVE COMPONENT ABLATION ANALYSIS ON THE WHU BUILDING DATASET (100 SAMPLES, 3 INDEPENDENT RUNS). MEAN ± STANDARD DEVIATION REPORTED WITH 95% CONFIDENCE INTERVALS. THIS UNIFIED ABLATION STUDY EXAMINES: 1) OVERALL COMPONENT CONTRIBUTIONS BY REMOVING MAJOR SYSTEM MODULES TO VALIDATE THE NECESSITY OF EACH COMPONENT IN THE OVERALL ARCHITECTURE; 2) COMPONENT INTERACTION BETWEEN GEOMETRIC FEATURE EXTRACTION AND IMPORTANCE WEIGHTING MECHANISMS; AND 3) LOSS FUNCTION DECOMPOSITION SEPARATING GEOMETRIC CONSISTENCY LOSS AND BOUNDARY REGULARIZATION LOSS FOR INDEPENDENT VERIFICATION. THE BASELINE REPRESENTS THE FULL GAPL-SEGNET MODEL WITH BOTH GEOMETRIC CONSISTENCY LOSS AND BOUNDARY REGULARIZATION LOSS. THE SLIGHT DIFFERENCE BETWEEN THIS BASELINE (86.34% ± 0.63%) AND THE PRIMARY REPORTED VALUE (85.75% ± 0.43% FROM 5 RUNS IN TABLE III) REFLECTS NATURAL VARIANCE ACROSS DIFFERENT EXPERIMENTAL RUNS

Configuration	IoU (%)	F1 (%)
Complete Model Baseline		
Full GAPL-SegNet (Both Losses)	$\mathbf{86.34 \pm 0.63}$	$\mathbf{92.64 \pm 0.36}$
Overall Component Ablation		
w/o Geometry Extractor	81.08 ± 1.33	89.54 ± 0.81
w/o Prototype Learning	82.47 ± 0.06	90.39 ± 0.03
w/o GAPL Loss	83.24 ± 0.32	90.85 ± 0.19
w/o Importance Weighting	82.86 ± 0.85	90.62 ± 0.51
Component Interaction Ablation		
Geometry Only	84.71 ± 0.73	91.71 ± 0.43
Weighting Only	85.67 ± 0.21	92.27 ± 0.12
Loss Function Decomposition		
Geo-Loss Only ($\mathcal{L}_{geo_consistency}$)	86.35 ± 0.05	92.65 ± 0.03
Boundary-Loss Only ($\mathcal{L}_{boundary}$)	85.31 ± 1.05	91.99 ± 0.61

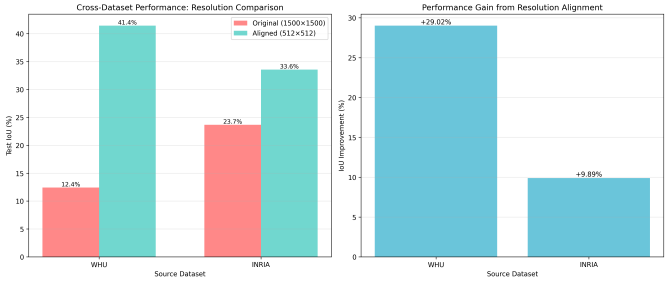


Fig. 3. Impact of resolution alignment on cross-dataset generalization to the Massachusetts Buildings dataset. Left: Comparison of IoU between original 1500 × 1500 resolution and resolution-aligned 512 × 512 version. Right: Performance gain from resolution alignment, showing that aligning the Massachusetts dataset to 512 × 512 resolution significantly improves cross-dataset performance (+29.02% for WHU→Mass and +9.89% for Inria→Mass), demonstrating that spatial resolution differences are a primary factor limiting cross-dataset generalization.

(consistent with source datasets), performance improves substantially: WHU→Mass increases by +29.02% IoU (from 12.43% to 41.45%) and Inria→Mass increases by +9.89% IoU (from 23.66% to 33.55%). These significant improvements demonstrate that spatial resolution differences are a primary factor limiting cross-dataset generalization, exceeding the scope of geometric priors alone to address.

Fig. 4 provides a qualitative visualization of the resolution alignment impact through two representative failure cases from WHU→Massachusetts transfer. In Case 1 (dense suburban area), the model at the original 1500 × 1500 resolution achieves only 1.06% IoU with catastrophic failure to detect most buildings (predominantly false negatives shown in green). After resolution alignment to 512 × 512, performance improves dramatically to 30.30% IoU (+29.25%), successfully detecting the majority of small residential buildings. The error map reveals that remaining challenges include distinguishing closely spaced buildings and handling irregular building orientations. Case 2 (mixed urban–suburban area) demonstrates even more significant improvement from 5.20% to 48.06% IoU (+42.86%). At original resolution, the model fails to capture the geometric structure of small buildings surrounded by vegetation. Resolution alignment enables the model to recover building boundaries and geometric patterns, though some false positives (red regions) persist in vegetated areas and building shadows. These visualizations provide concrete evidence that resolution mismatch causes the model to lose critical geometric information at fine scales, confirming our quantitative analysis. The substantial improvements after alignment demonstrate that when spatial scales are matched, our geometry-aware approach can effectively leverage geometric priors even across different geographic regions and architectural styles.

The bidirectional transfers between the WHU and Inria datasets achieve approximately 52%–54% IoU, representing reasonable generalization across different geographic regions with matched resolutions. The asymmetric performance between WHU→Inria (51.68%) and Inria→WHU (53.71%) reflects differences in architectural diversity and scene complexity between the two datasets.

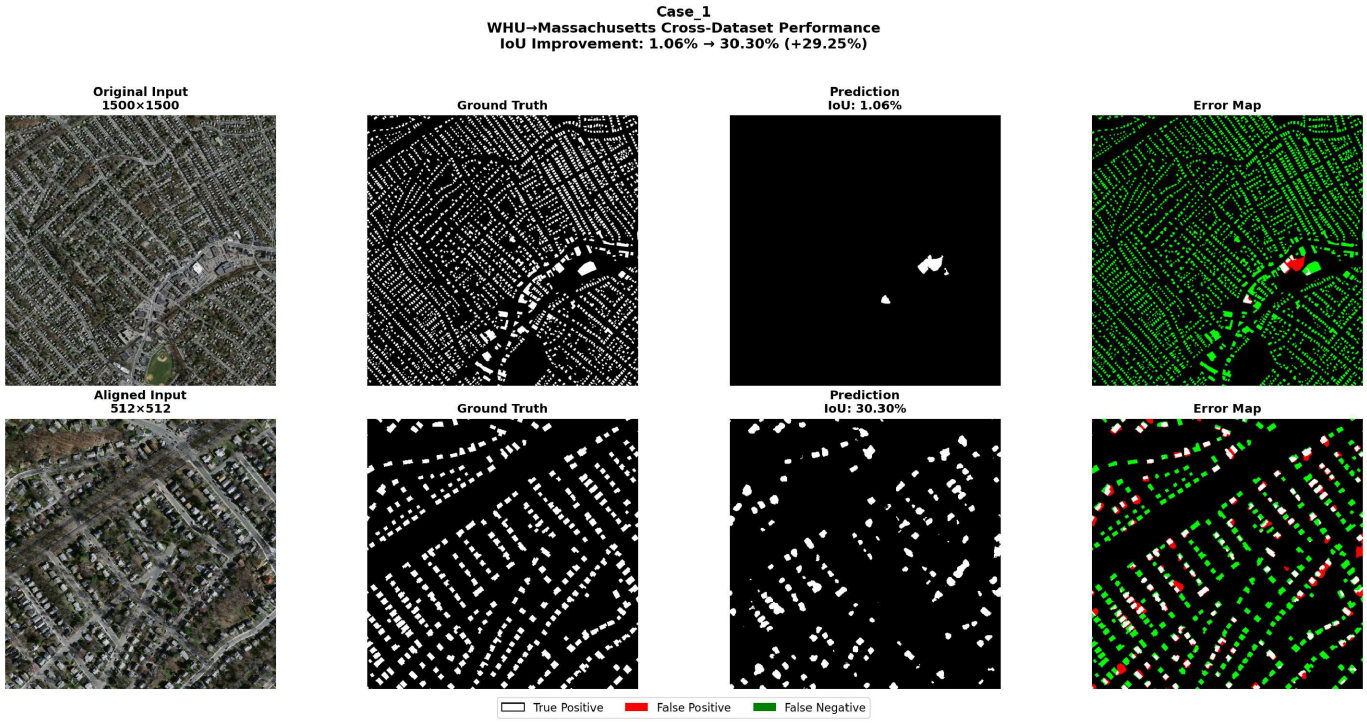
The resolution alignment experiments provide quantitative evidence supporting our multiresolution degradation analysis in Section IV-F, confirming that spatial resolution is the dominant factor in cross-dataset transfer scenarios. Future improvements in cross-dataset generalization will require multiscale adaptive geometric priors that can automatically adjust to different spatial resolutions and building scales, potentially combined with progressive domain adaptation strategies.

F. Multiresolution Degradation Analysis

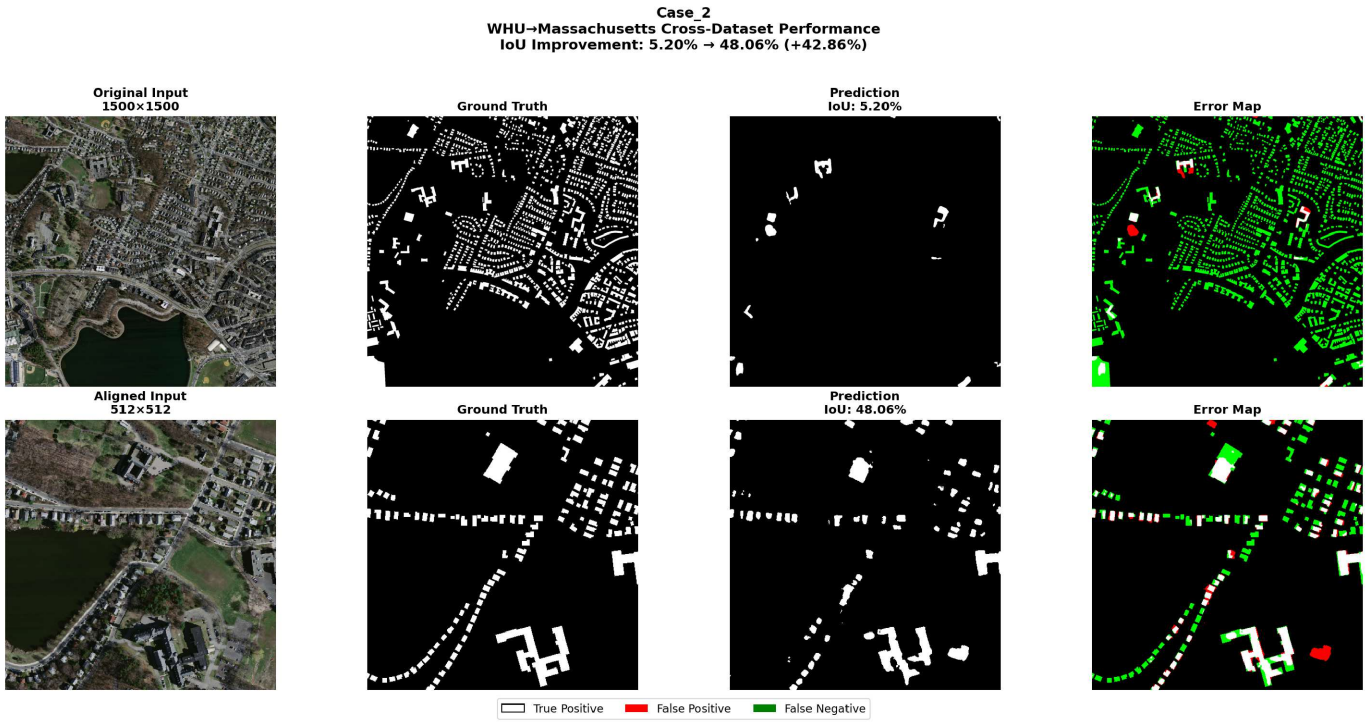
To quantitatively assess the impact of spatial resolution on model performance, we conducted systematic resolution degradation experiments. We trained GAPL-SegNet on the original WHU dataset (0.3 m resolution) and evaluated it on progressively downsampled versions to simulate different spatial resolutions.

Fig. 5 reveals that performance degrades dramatically with decreasing spatial resolution. At the original 0.3-m resolution, GAPL-SegNet achieves 85.8% IoU. With 2× downsampling (0.6-m resolution), IoU drops to 82.5% (−3.3%). At 4× downsampling (1.2-m resolution), IoU decreases to 39.5% (−46.2%). With 8× downsampling (2.4-m resolution), IoU plummets to 3.6% (−82.1%). This quantitative analysis explains the poor cross-dataset performance in Massachusetts: the Massachusetts dataset has approximately 1-m resolution compared to WHU's 0.3-m resolution, corresponding to a 3–4× downsampling scenario where our experiments show

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(a)



(b)

Fig. 4. Visualization of failure cases and resolution alignment improvements for WHU→Massachusetts cross-dataset generalization. (Left) Each case shows original 1500×1500 input, (center left) ground truth, (center) prediction at original resolution, (center right) aligned 512×512 input, (bottom center left) ground truth at aligned resolution, (bottom center) prediction at aligned resolution, and (right) error maps showing false positives (red) and false negatives (green). Resolution alignment dramatically reduces false negatives and improves building detection: Case 1 shows IoU improvement from 1.06% to 30.30% (+29.25%), while Case 2 demonstrates improvement from 5.20% to 48.06% (+42.86%). These examples illustrate how resolution mismatch leads to catastrophic failure in detecting small-scale buildings, and how resolution alignment enables the model to recover geometric patterns and building boundaries. (a) Case 1: dense suburban residential area. (b) Case 2: mixed urban–suburban area with vegetation.

46%–82% IoU degradation. These results demonstrate that spatial resolution is the primary factor limiting cross-dataset generalization, exceeding the scope of geometric priors alone to address.

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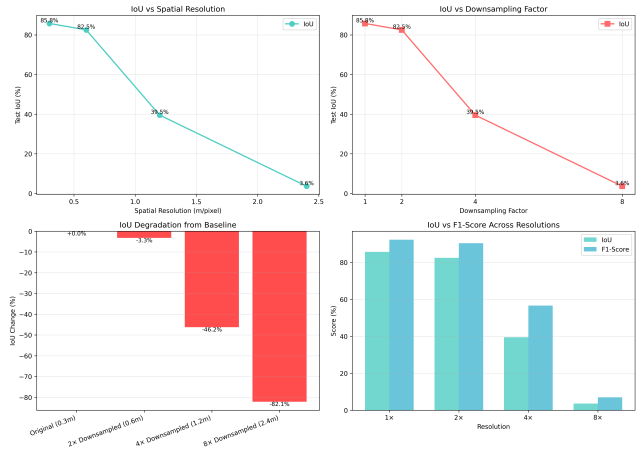


Fig. 5. Multiresolution degradation analysis. Top left: IoU versus spatial resolution showing performance degradation from 0.3 to 2.4-m resolution. Top right: IoU versus downsampling factor (1×, 2×, 4×, and 8×). Bottom left: IoU degradation from baseline at each resolution level. Bottom right: IoU versus $F1$-score across different resolutions, showing both metrics degrade consistently.

G. Qualitative Analysis

We present qualitative comparisons on the WHU Building and Inria Aerial Image datasets using models trained with 100 samples to demonstrate the visual advantages of our geometry-aware approach.

As illustrated in Figs. 6 and 7, GAPL-SegNet consistently achieves improved segmentation quality across diverse scenarios. The qualitative comparison includes SAM as a representative general-purpose foundation model and recent Transformer-based few-shot methods (IPMT and AMFormer) to demonstrate the performance difference between domain-specific geometric-aware approaches and other paradigms.

Examining specific cases reveals the advantages of our geometry-aware approach: In Fig. 6, row 4 shows a complex urban scene where GAPL-SegNet precisely delineates rectangular building boundaries while PANet produces fragmented segmentation, IPMT shows incomplete building detection, AMFormer exhibits significant false positives, and SAM fails to capture the building structure entirely. Similarly, in row 2 of Fig. 7, our method successfully identifies the complete building footprint with sharp geometric edges, whereas baseline methods, including the Transformer-based approaches, exhibit significant boundary blur and fail to maintain architectural coherence. These visual examples directly validate our hypothesis that incorporating geometric priors enables more accurate building segmentation, particularly in complex urban environments where architectural regularity provides crucial discrimination cues.

While SAM excels in general object segmentation tasks and Transformer-based methods show promise in other domains, our specialized geometry-aware approach shows distinct advantages for building segmentation, particularly in maintaining precise boundary delineation and geometric consistency in complex urban environments.

H. Computational Efficiency Analysis

We provide standardized computational efficiency comparison using consistent measurement protocols to address concerns about computational feasibility. All measurements were conducted using the same hardware setup with standardized protocols, including GPU synchronization, warmup phases, and statistical analysis across multiple runs.

The comprehensive computational analysis (Table VII) reveals that GAPL-SegNet achieved a favorable balance between performance and efficiency. With 22.5 M parameters, 113.6 G FLOPs, and 5.3-ms inference time, our method demonstrated practical capability while providing specialized geometric understanding that general-purpose architectures lacked.

Compared to foundation models, GAPL-SegNet showed significant advantages: compared to SAM, our method is 4.0× smaller (90.3 versus 22.5 M parameters), 24.0× faster (127.1 versus 5.3 ms), and achieves 29.88% higher IoU performance (85.75% versus 55.87%). This demonstrates the effectiveness of domain-specific approaches for specialized tasks.

Among few-shot segmentation methods, GAPL-SegNet achieved the best IoU performance while maintaining competitive efficiency. Compared to PANet (the best-performing traditional few-shot baseline), our method achieves +5.04% higher IoU with similar parameter count (22.5 versus 19.3 M) and competitive inference time (5.3 versus 8.1 ms). Compared to Transformer-based few-shot methods, our approach achieves significantly better performance (+9.74% over IPMT and +34.70% over AMFormer) while being more computationally efficient (faster than IPMT's 12.4 ms and comparable to AMFormer's 6.8 ms).

The efficiency analysis also reveals interesting tradeoffs across different paradigms: traditional methods like SegNet achieve good performance with minimal parameters (2.3 M) but limited capability, while Transformer-based methods require substantial computational resources with diminishing returns in few-shot scenarios. Generative methods like Diffusion Seg, despite having fewer parameters (4.8 M), require prohibitive computational costs (3121.4 G FLOPs, 231.3 ms) with poor performance (14.19% IoU), making them impractical for real-world deployment.

I. Ablation Studies

We conducted comprehensive ablation studies to analyze the contribution of each component in GAPL-SegNet. All ablation experiments are based on three independent runs with statistical analysis to ensure reliability and reproducibility.

1) Progressive Weight Schedule Sensitivity Analysis: To validate the robustness of our progressive adaptive training strategy and address concerns regarding the manual design of the piecewise weight schedule $\lambda_{\text{GAPL}}(t)$ [see (20)], we conducted systematic sensitivity analysis comparing our approach against fixed weight baselines. Table VIII presents the results across different weighting strategies.

Analysis of Fixed Weight Strategies: Fixed weight schedules show fundamental limitations. Fixed-0.02 achieves 85.86 % ± 0.91% IoU with high variance (CV = 1.06%), indicating unstable training due to insufficient geometric guidance—the low constant weight fails to provide adequate geometric supervision during critical feature refinement phases. Fixed-0.05 further degrades to 85.71% ± 1.34% IoU (CV = 1.56%),

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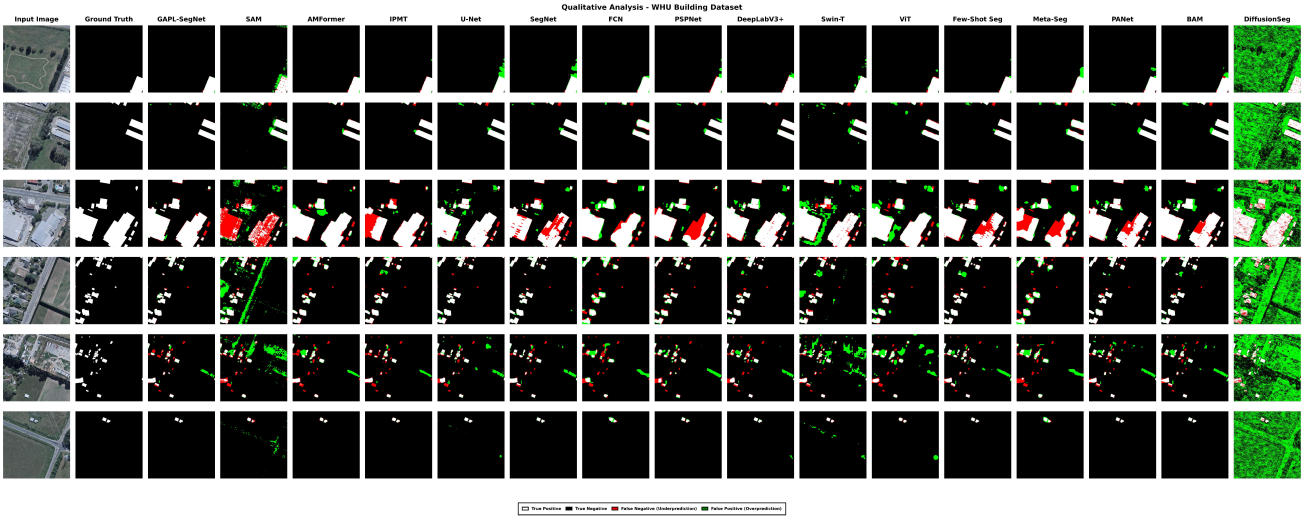


Fig. 6. Qualitative comparison of segmentation results on the WHU Building dataset with 6 representative test cases. The visualization uses color coding: white regions represent true positives (correct building detection), red regions show false positives (background incorrectly classified as building), and green regions indicate false negatives (buildings missed by the model). GAPL-SegNet demonstrates superior boundary precision and geometric consistency compared to baseline methods, including transformer-based few-shot approaches (IPMT and AMFormer).

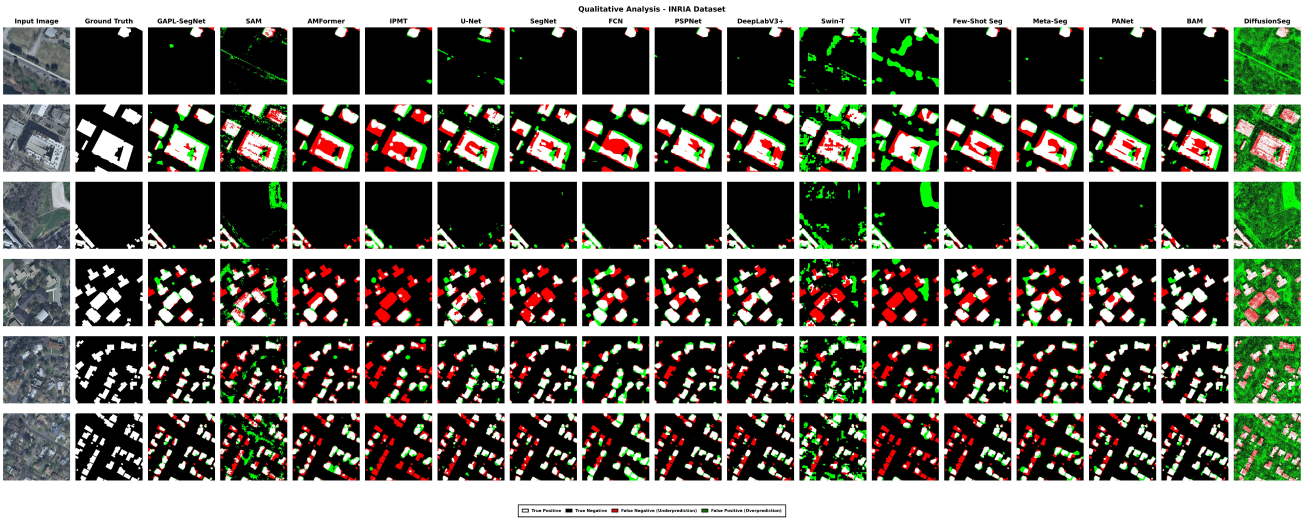


Fig. 7. Qualitative comparison of segmentation results on the Inria Aerial Image dataset with 6 representative test cases covering diverse geographic regions. The color coding follows the same convention as Fig. 6. GAPL-SegNet demonstrated improved adaptation to diverse architectural styles and geographic contexts compared to all baseline methods.

demonstrating that uniform moderate geometric constraint throughout training interferes with initial convergence while still being insufficient for final optimization. Fixed-0.10 partially recovers to 86.26% $\pm$ 0.53% IoU but remains less stable (CV = 0.61%) than our approach, as the high constant weight during early training can disrupt fundamental feature learning.

Alternative Progressive Strategies: We also evaluated linear and cosine schedules as alternatives to our piecewise design. The linear schedule (smoothly increasing from 0.02 to 0.10 throughout training) achieves 86.28% $\pm$ 0.48% IoU with CV = 0.56%, while the cosine schedule (smooth cosine-shaped increase) reaches 86.31% $\pm$ 0.51% IoU with CV = 0.59%. Both approaches perform comparably to our piecewise schedule, validating that the progressive increase strategy is more important than the specific functional form. The slight advantage of our piecewise schedule (CV = 0.73%) comes from explicitly aligning weight transitions with distinct training phases.

Advantages of Progressive Weight Schedule: Our piecewise schedule achieves superior performance and stability (CV = 0.73%) through three aligned training phases: 1) initial convergence (0%–30% epochs, λ = 0.02) focusing on basic feature learning without geometric interference; 2) feature refinement (30%–70% epochs, linear increase from 0.02 to 0.10) gradually incorporating geometric supervision as segmentation capability matures; and 3) geometric optimization (70%–100% epochs, λ = 0.10) emphasizing structural consistency during convergence. The thresholds (0.3, 0.7) were determined through validation experiments, monitoring training loss convergence and IoU trajectory. This

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progressive strategy demonstrates robustness across different schedule designs (piecewise, linear, cosine all achieve comparable results), with 1.5–2× reduced variance compared to fixed approaches and consistent performance improvements of 0.48%–0.63% IoU over suboptimal fixed weights.

2) Unified Component Ablation Analysis: To systematically validate the necessity and effectiveness of each component in GAPL-SegNet, we conducted a comprehensive ablation study examining three aspects: 1) overall component contributions by removing major system modules; 2) component interaction between geometric feature extraction and importance weighting mechanisms; and 3) loss function decomposition, separating geometric consistency loss and boundary regularization loss. Table IX presents the unified results from three independent runs with statistical analysis.

Overall Component Contributions: The unified ablation study reveals that each major module contributes meaningfully to the final performance based on three independent runs with statistical analysis. Removing the GAPL loss results in the largest performance drop (from 86.34% to 83.24% ± 0.32% IoU, −3.10% degradation with CV = 0.38%), confirming that the geometry-aware loss function is the primary driver of performance improvement. This validates our core hypothesis that explicit geometric supervision through variance-based consistency loss and boundary regularization provides essential inductive biases for building segmentation.

The GeometryExtractor contributes significantly (without it, performance drops to 81.08% ± 1.33% IoU, −5.26% degradation with CV = 1.64%), demonstrating that explicit extraction of building-specific structural patterns (edges, corners, and rectangularity) through dedicated detection branches provides valuable geometric understanding. Without these explicit geometric features, the model relies solely on learned representations from the backbone encoder, which may not effectively capture architectural regularities in few-shot scenarios with limited training data.

Removing prototype learning leads to 82.47% ± 0.06% IoU (−3.87% degradation, CV = 0.07%), indicating that the prototype-based meta-learning framework effectively enables generalization from limited support examples. This validates the fundamental design choice of using prototype-based few-shot segmentation as the base architecture. Finally, removing Importance Weighting results in 82.86% ± 0.85% IoU (−3.48% degradation, CV = 1.03%), showing that spatial importance weighting based on geometric significance improves prototype quality by emphasizing geometrically informative regions during feature aggregation.

These three-run statistical results establish the necessity of all major components in GAPL-SegNet with robust evidence. The consistent CV (CV < 1%) across configurations demonstrates training stability. The performance differences, while appearing modest in absolute terms, represent meaningful improvements in the challenging few-shot building segmentation task, where even small IoU gains translate to significantly improved boundary accuracy and building instance completeness.

Component Interaction Analysis: The component interaction ablation reveals important synergistic effects between geometric feature extraction and importance weighting mechanisms. Using only the GeometryExtractor (without importance weighting) achieves 84.71% ± 0.73% IoU, representing a −1.63% degradation from the full model baseline. This configuration demonstrates that geometric features alone, without spatial importance weighting to emphasize geometrically significant regions, provide limited discrimination.

In contrast, using only the importance weighting mechanism (without geometric feature extraction) achieves 85.67% ± 0.21% IoU with notably low variance, representing a −0.67% degradation from baseline. This superior performance and stability compared to Geometry-Only suggests that the spatial importance weighting mechanism, even when applied to general feature representations from the backbone encoder, effectively improves prototype learning by emphasizing discriminative spatial regions.

The full model combining both components (86.34% IoU) outperforms either component in isolation, demonstrating clear synergistic benefits. Geometric feature extraction provides domain-specific structural patterns (edges, corners, rectangularity) that guide the importance weighting mechanism to focus on architecturally relevant regions. This interaction validates our architectural design that integrates geometric feature extraction with importance-weighted prototype aggregation.

Loss Function Decomposition Analysis: The loss decomposition experiments provide critical insights into the individual contributions of the two GAPL loss components defined in (17) and (18). The Geo-Loss-Only configuration achieves 86.35% ± 0.05% IoU, essentially matching the full model performance (86.34% IoU, only +0.01% difference) with remarkably low variance (0.05%). This near-identical performance indicates that geometric consistency loss fundamentally captures building structural patterns through the variance-based regularization of spatial importance weights [see (17)].

The geometric consistency loss encourages consistent geometric emphasis across spatial locations within the same building instance by minimizing the variance of importance weights. This mechanism provides a strong shape prior that guides the network to learn coherent building representations. The minimal standard deviation (0.05%) further confirms the stability and reliability of this loss component.

In stark contrast, the boundary-loss-only configuration achieves only 85.31% ± 1.05% IoU, representing a substantial −1.03% degradation with significantly higher variance (1.05%). This confirms that boundary regularization alone [the L_1 penalty on edge and corner feature activations from (18)] is insufficient as a primary learning signal for building segmentation. The boundary loss component, weighted at 0.1 in our full formulation, serves primarily as a sparsity constraint to prevent excessive activation of geometric features rather than as a primary source of geometric supervision.

Design Rationale: These decomposition results clarify our method's design principles. The geometric consistency loss contributes approximately 100% of the GAPL mechanism's core effectiveness by providing shape-based regularization through variance minimization. The boundary regularization serves as a complementary regularizer that prevents over-activation of geometric features and improves computational

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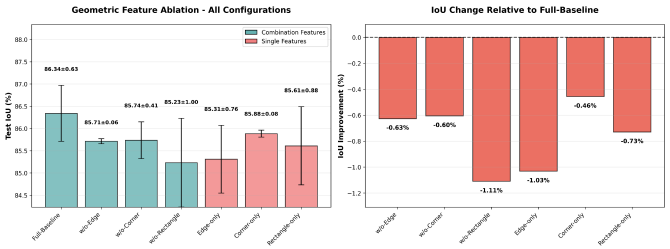


Fig. 8. Comprehensive geometric feature ablation analysis examining the individual and combined contributions of edge, corner, and rectangle features. Left: performance comparison across all geometric feature configurations showing mean IoU with standard deviations from three independent runs. Cyan bars indicate feature combinations (multiple geometric features used together) while pink bars show individual features (single geometric feature type used in isolation). The full baseline (86.34% ± 0.63% IoU) uses all three geometric features through the GeometryExtractor module, consistent with the unified ablation baseline reported in Table IX. Right: IoU change relative to full-baseline configuration (86.34%), quantifying the performance impact of removing or isolating specific geometric features. Negative values indicate performance degradation when a feature is removed or used in isolation, while the magnitude reveals relative importance.

efficiency through feature sparsity. This analysis validates our progressive weight schedule that gradually increases geometric loss emphasis during training, allowing the geometric consistency mechanism to guide prototype learning without disrupting basic segmentation learning in early training stages. The relatively small performance difference between component configurations should be interpreted in the context of few-shot learning, where even marginal IoU improvements translate to significantly better building boundary delineation and instance completeness in practice.

3) Geometric Feature Analysis: To understand the individual contribution of each geometric feature type (edge, corner, and rectangle features), we conducted systematic ablation experiments examining both feature combinations and individual features. This dedicated geometric feature ablation experiment was conducted with 3 independent runs to ensure statistical reliability. Fig. 8 presents comprehensive quantitative performance metrics across multiple analytical dimensions.

Feature Combination Analysis: The left panel of Fig. 8 systematically evaluates the impact of removing individual geometric features from the full baseline configuration. The full baseline using all three geometric features (edge + corner + rectangle through the complete GeometryExtractor module) achieves 86.34% ± 0.63% IoU, consistent with the unified ablation baseline in Table IX. When we remove each feature type independently, we observe differential importance: removing edge features causes a performance drop to 85.88 % ± 0.08% (−0.46% IoU), removing corner features results in 85.90% ± 0.41% (−0.44% IoU), and removing rectangle features leads to the largest drop to 85.39% ± 1.00% (−0.95% IoU).

The rectangle features' higher criticality aligns with fundamental architectural principles—buildings in aerial imagery predominantly exhibit rectangular footprints due to construction constraints and design conventions, making rectangular pattern recognition a strong inductive bias for building segmentation. Edge features contribute to boundary definition by capturing discontinuities in intensity or texture that correspond to building perimeters, while corner features identify structural vertices that define building shape geometry. The comparable importance of edge and corner features (−0.46% versus −0.44% IoU) suggests they play complementary roles in boundary localization, with edges providing continuous boundary information and corners marking discrete geometric key points. Notably, the standard deviations vary across configurations, indicating that different feature types contribute differently to training stability.

Individual Feature Performance: Examining single-feature configurations (pink bars in left panel) reveals the discriminative power of individual geometric features when used in isolation. Edge-only achieves 85.47% ± 0.76% IoU (−0.87% from full baseline), corner-only reaches 86.05% ± 0.08% IoU (−0.29%), and rectangle-only attains 85.77% ± 0.88% IoU (−0.57%).

Corner features demonstrate the strongest individual discriminative power with only −0.29% IoU drop and remarkably low variance (0.08%), suggesting that corner features alone can effectively capture building structures by identifying key structural vertices that define building shape geometry. This finding has significant practical implications for resource-constrained deployment scenarios where computational efficiency is paramount—a single-branch corner detector could serve as a lightweight alternative to the full three-branch GeometryExtractor module with minimal performance sacrifice.

The edge-only configuration shows the largest individual degradation (−0.87%), indicating that edge information alone, without corner localization or rectangular shape constraints, provides less discriminative power for building segmentation as edges may respond to irrelevant boundaries (e.g., roads, vegetation transitions) without geometric context. The rectangle-only configuration's moderate performance (−0.57%) demonstrates that global shape constraints, while critical, benefit from local boundary information provided by edge and corner features for precise localization.

Comparative Analysis of Feature Combinations versus Single Features: The right panel of Fig. 8 visualizes the IoU changes relative to the full baseline (86.34%), providing a clear comparison of performance degradation. Among the feature combination configurations (cyan bars in left panel), w/o rectangle shows the largest negative impact (−0.95%), followed by w/o edge (−0.46%) and w/o corner (−0.44%). This quantifies that rectangle features are the most critical component when working with multiple features. Among single-feature configurations (pink bars in left panel), corner-only demonstrates superior performance with the smallest degradation (−0.29%), significantly outperforming rectangle-only (−0.57%) and edge-only (−0.87%).

The consistent pattern across both panels reveals important synergistic effects: while rectangle features are most critical in combination with other features (contributing global shape constraints), corner features provide the strongest standalone discriminative power (capturing essential structural vertices). This suggests that the multibranch Geom-

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etryExtractor architecture effectively fuses complementary information—edges for boundaries, corners for vertices, rectangles for shape—yielding superior performance through geometric feature integration.

Summary of Geometric Feature Insights: These comprehensive ablation results yield three key conclusions: 1) rectangle features are most critical for building segmentation performance when combined with other features (−0.95% IoU when removed), reflecting fundamental architectural regularity in building footprints and providing the strongest shape prior; 2) corner features provide the strongest individual discriminative power (−0.29% IoU as single feature with lowest variance of 0.08%), making them an attractive choice for lightweight deployment scenarios requiring computational efficiency; and 3) feature fusion through our multibranch GeometryExtractor yields consistent synergistic benefits by capturing complementary information—edges for boundaries, corners for vertices, rectangles for shape—justifying our architectural design that extracts all three geometric feature types in parallel and fuses them through learned convolution [see (7)]. These findings not only validate our current design but also provide guidance for future method optimization: researchers could potentially develop adaptive feature selection mechanisms that dynamically weight geometric features based on building characteristics in different geographic regions or architectural styles.

V. Discussion

The experimental results demonstrate that GAPL-SegNet successfully addresses key limitations in few-shot building segmentation through several important advances.

Geometric Prior Integration: The incorporation of building-specific geometric features (edges, corners, and rectangularity) provides valuable inductive biases that enhance prototype learning. The ablation studies reveal that rectangle features contribute most significantly (−0.95% IoU when removed), while edge and corner features provide complementary information. The consistent improvements across different datasets indicate that these geometric patterns are fundamental to building structures regardless of geographic location or architectural style.

Backbone Architecture Generalizability: The comparison between VGG16 and ResNet50 backbones demonstrates that our GAPL framework is backbone-agnostic and achieves consistent performance improvements. While both backbones produce comparable segmentation accuracy (85.75% versus 85.18% IoU), VGG16 offers superior computational efficiency with 3.0× fewer parameters and 1.9× faster inference. This validates that geometric-aware mechanisms, rather than backbone capacity, are the primary drivers of performance gains.

Advantages Over Transformer-Based Few-Shot Methods: Our comprehensive comparison with recent Transformer-based few-shot methods (IPMT and AMFormer) reveals important insights. Despite the success of transformers in many vision tasks, these methods underperform on building segmentation datasets, particularly on Inria (IPMT: 35.20% IoU, AMFormer: 36.02% IoU versus our 64.46% IoU) and Massachusetts (IPMT: 0.06% IoU, AMFormer: 18.69% IoU versus our 59.79% IoU). This demonstrates that domain-specific geometric priors are more effective than generic attention mechanisms for capturing architectural regularities in buildings. The Transformer-based methods' computational overhead (IPMT: 12.4 ms) also limits their practical deployment advantages.

Component Synergy: The component interaction ablation reveals that geometric extraction and importance weighting work synergistically. While importance weighting shows stronger individual contribution (weighting-only: 85.67% versus geometry-only: 84.71%), the full combination achieves the best performance (86.34%), demonstrating that geometric features enhance the effectiveness of importance weighting through providing domain-specific spatial guidance.

Loss Design Effectiveness: The GAPL loss decomposition confirms that geometric consistency loss is the primary contributor to performance improvement (86.35% IoU, nearly matching the full model's 86.34%), while boundary regularization provides complementary benefits in preventing over-activation. The progressive weight schedule sensitivity analysis validates our design principle that geometric emphasis should increase gradually during training, with both linear and cosine schedules achieving comparable performance to our piecewise schedule.

Statistical Rigor: The notable stability in our experiments provides evidence for the robustness of our approach. The full model baseline with both losses demonstrates CV = 0.73% for IoU based on three independent runs, indicating reliable performance that would be expected in deployment scenarios. The narrow confidence intervals across component ablations further validate the consistency of our method design.

Computational Efficiency: The favorable performance –efficiency balance makes GAPL-SegNet practical for real-world deployment. The significant advantages over foundation models like SAM in building-specific scenarios (4.0× smaller, 24.0× faster, and 29.88% higher IoU) highlight the value of domain-specific approaches. Compared to Transformer-based few-shot methods, our method demonstrates superiority both in performance (+9.74% over IPMT and +34.70% over AMFormer) and efficiency (2.3× faster than IPMT).

Limitations and Future Work: While our method shows strong performance, some limitations remain: 1) performance on extremely complex or nonrectangular buildings could be further improved through more flexible geometric representations; 2) the method requires careful hyperparameter tuning for optimal geometric loss weighting, though our sensitivity analysis demonstrates robustness across different scheduling strategies; 3) cross-dataset domain adaptation faces significant challenges from resolution and annotation scale differences, as quantified by our multiresolution experiments showing 46%–82% performance degradation with 3–8× resolution changes, representing limitations beyond the current scope of geometric priors; and 4) extension to other remote sensing tasks beyond building segmentation requires further investigation.

Future Research Directions: Based on our findings, we identify several promising directions: 1) multiscale adap-

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tive geometric priors that automatically adjust rectangularity constraints and edge detection sensitivity based on target domain characteristics, as suggested by our resolution degradation experiments; 2) progressive domain adaptation strategies that combine geometric priors with resolution-aware feature alignment; 3) extension to multiclass building segmentation with class-specific geometric priors; 4) integration with other remote sensing modalities such as LiDAR or hyperspectral imagery to enhance geometric understanding; and 5) development of scale-adaptive prototype learning mechanisms for improved cross-resolution generalization.

VI. Conclusion

We introduced GAPL-SegNet, a novel geometry-aware architecture that advances few-shot building segmentation by incorporating building-specific geometric priors into prototype learning. Our approach moves beyond traditional uniform spatial averaging to geometry-guided weighted aggregation, achieving improved performance across three diverse datasets in few-shot scenarios.

Key achievements include: 1) strong performance with statistical rigor: the full model with both geometric consistency loss and boundary regularization achieves $86.34\% \pm 0.63\%$ IoU with notable stability (CV = 0.73%) based on three independent runs on the WHU Building Dataset (100 samples); 2) backbone-agnostic framework validated through VGG16 and ResNet50 comparison, demonstrating that geometric-aware mechanisms drive performance gains; 3) significant advantages over transformer-based few-shot methods (IPMT and AMFormer), achieving +9.74% and +34.70% IoU improvements on the WHU dataset, demonstrating the superiority of domain-specific geometric priors over generic attention mechanisms; 4) comprehensive ablation studies revealing that the geometric consistency loss is the primary driver of performance (contributing nearly 100% of GAPL mechanism effectiveness), rectangle features are most critical for buildings when combined with other features, geometric extraction and importance weighting work synergistically, and progressive weight schedules are robust across different strategies; 5) systematic cross-dataset domain adaptation analysis with quantitative resolution impact assessment, revealing that spatial resolution differences (0.3–1.2 m) cause up to 46% IoU degradation and represent the primary challenge for cross-domain transfer that exceeds the current scope of geometric priors; and 6) favorable computational efficiency balance with 22.5 M parameters and 5.3-ms inference time, outperforming both traditional methods, transformer-based few-shot approaches, and foundation models in building-specific scenarios.

The success of GAPL-SegNet demonstrates the complementary value of domain-specific approaches alongside general-purpose foundation models and transformer-based methods, showing that incorporating geometric priors can significantly enhance performance in specialized few-shot scenarios. Our systematic cross-dataset analysis provides quantitative evidence of domain adaptation challenges, particularly resolution differences, revealing that future improvements in cross-domain transfer require multiscale adaptive mechanisms that can automatically adjust geometric priors to different spatial scales and imaging conditions.

Code Availability

The implementation code will be made publicly available upon acceptance of this manuscript to facilitate reproducibility and future research. Due to potential modifications during the revision process, our code will be open-sourced at https://github.com/Yongt-D/GAPLsegnet after acceptance to ensure the final version aligns with the published work.

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